

Econ 646 Spring 2012
Exam 1 (March 19th)

Our model economy is populated by identical agents who maximize their lifetime utility over sequences of consumption (C_t) and hours worked (N_t):

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t (\log C_t - N_t) \right]$$

Output (Y_t) is produced with a Cobb-Douglas production function using capital (K_t) and labor:

$$Y_t = K_t^\alpha N_t^{1-\alpha} \quad (1)$$

Output can be used for consumption or investment (I_t):

$$Y_t = C_t + \frac{I_t}{z_t} \quad (2)$$

Capital is accumulated as follows:

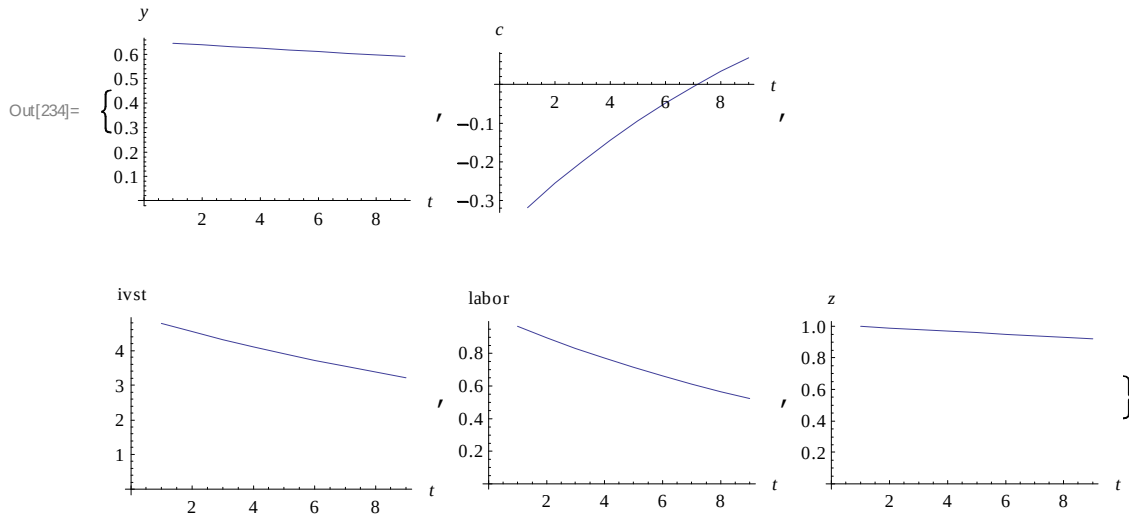
$$K_{t+1} = (1 - \delta) K_t + I_t \quad (3)$$

z_t is an exogenous random variable, following a stationary AR(1) process

$$\log z_t = \rho \log z_{t-1} + \varepsilon_t$$

z_t represents the current state of technology for producing capital goods, and an increase in z_t is interpreted as investment-specific technological progress. Answer the following questions.

1. Derive the first order conditions.
2. Find the steady state values of K/Y , I/Y , C/Y , and N .
3. Log-linearize the first order conditions and the technical constraints (1) to (3) around the steady state.
4. Figure 1 plots impulse response functions to a positive innovation in z_t under standard parameterization $(\beta, \alpha, \delta, \rho) = (0.99, 0.33, 0.025, 0.99)$. The shock hits in period 1, and percentage deviations from the steady state values are plotted. Explain why consumption drops on impact, and why output, labor hours, and investment rise.



Now, let's modify the model economy in the following ways. The production function is

$$Y_t = (u_t K_t)^\alpha N_t^{1-\alpha}$$

where u_t is the rate of capital utilization: we let the agents choose how intensively they use K_t . The cost of intensive capital usage is fast depreciation. Namely, the capital accumulation rule is modified as follows:

$$K_{t+1} = (1 - d(u_t)) K_t + I_t$$

where $d(u_t)$ is a convex function in the rate of utilization: $d'(u_t) > 0$, $d''(u_t) \geq 0$. The steady state capital depreciation is δ : $d(u) = \delta$. All the other parts are the same as before. You work with this modified model for the following 5 questions.

5. Derive the first order conditions.
6. Find the steady state values of K/Y , I/Y , C/Y , N , and $d'(u)u$.
7. Log-linearize the system of equations and the technical constraints around the steady state.
8. Show that the elasticity of $d'(u_t)$ evaluated in the steady state is the only new parameter that has to be calibrated to study the log-linearized system.
(HINT: the elasticity of $d'(u_t)$ evaluated in the steady state is $d''(u)u/d'(u)$. What you need to do is showing only five parameters $(\beta, \alpha, \delta, \rho, d''(u)u/d'(u))$ appears in the log-linearized system.)
9. Figure 2 plots impulse response functions to a positive innovation in z_t under the following parameterization $(\beta, \alpha, \delta, \rho, d''(u)u/d'(u)) = (0.99, 0.33, 0.025, 0.99, 0.25)$. Explain (i) why the rate of utilization rises on impact, (ii) why consumption rises on impact, and (iii) why output, investment and hours worked move much larger than those in Figure 1 although the size of the shock is the same.

